

Downscaling stratospheric aerosol injection simulations: Convenings summary

The goal of this project was to produce a high-resolution version of global stratospheric aerosol injection (SAI) scenarios to enable research on potential regional impacts on domains such as public health, agriculture, and ecosystems — particularly in the global South — and an exploration of the uncertainties introduced by downscaling itself. While the dataset is fully public, its ultimate utility depends on how responsive our technical work is to the communities who may be affected by it. To that end, we conducted a series of convenings at the start of the project to gather input from communities whose expertise could inform the design of a usable and useful dataset.

In 2025, we held a workshop at the Geoengineering Model Intercomparison Project (GEOMIP). Building on this engagement, we conducted six, 1.5-hour convenings in September and October 2025 with the following groups: civil society, downscaling experts, climate modelers, impact modelers, technical experts unaffiliated with the Degrees Initiative or GEOMIP, and social scientists. All convenings were conducted under Chatham House Rules to encourage open discussion. In total, 122 individuals participated, representing 27 organizations across nine countries.

This document summarizes the discussions from these convenings, organized by the key implementation decisions for this project:

1. Which SAI scenarios to downscale
2. Which models to downscale and how many ensemble members to use
3. Which variables to downscale
4. Which bias correction and downscaling algorithm to use
5. Which observational dataset to use
6. How to effectively communicate uncertainties
7. How to ensure appropriate access to the downscaled dataset

For each decision category, we highlight areas of convergence and divergence in the recommendations made across expert groups, and describe how the convenings informed our ultimate project design and implementation choices.

01 — Which SAI scenarios to downscale

Stratospheric aerosol injection (SAI) could be deployed in many ways, represented in climate models by a range of scenarios that vary in the global emissions trajectory, how much SAI is deployed (e.g., to hit a temperature target), and what injection strategy is used. Based on initial research and input at the GEOMIP convening, we entered the convenings with a tentative proposal to use the G6-1.5K SAI scenario, which is designed to stabilize global warming at 1.5 degrees above preindustrial levels and is based on SSP2-4.5, a medium emissions pathway.

No red flags were raised around moving forward with the G6-1.5K simulations, but there was a strong consensus across convenings that downscaling additional SAI scenarios would help illustrate the uncertainty inherent in representing the multitude of possible futures within climate models (*civil society, climate modelers, impact modelers, and technical experts convenings*). Participants differed on which additional scenarios would be most useful. Several suggested choosing scenarios with maximal differences to highlight design uncertainty (*climate modelers, impact modelers, and technical experts convenings*). Others cautioned that maximizing differences could spotlight unrealistic futures, emphasizing that selecting a scenario implicitly signals that it's worthwhile to continue exploring and learning about that scenario (*climate modelers convening*). Similar recommendations stressed the value of politically possible scenarios over “model-kicking” ones (*GEOMIP and technical experts convening*), including examples such as limited deployment, non-coordinated or non-ideal trajectories, and scenarios in which background warming continues. Specific scenarios, such as GLENS, ARISE-SAI-1.5, and HiLLa, were also proposed for consideration (*GEOMIP and downscaling experts convenings*).

We decided to move forward with the G6-1.5K scenario due to its recent development, strong scientific and policy relevance, prior multi-model runs, and planned inclusion in CMIP7. We also decided to design and run a termination shock scenario in order to illustrate the power of scenario choice, and enable impact research into a non-ideal deployment scenario.

02 — Which models and ensemble members to downscale

Climate models differ in structure and assumptions, so the same SAI scenario can produce different outcomes across models. Even within a single model, different initial conditions or assumptions can yield varying results — captured by multiple model runs called ensemble members. Downscaling the same scenario across different models and ensemble members helps characterize the uncertainty inherent in climate model projections. The choice of climate

models and ensemble members is closely tied to the choice of SAI scenarios, since not all scenarios have been run on all climate models.

Participants generally agreed that using multiple models and ensemble members was critical (*GEOMIP, climate modelers, impact modelers, and technical experts convenings*), but highlighted that more work is needed to determine which sources of model uncertainty are most consequential for understanding SAI impacts across different regions and timescales (*climate modelers convening*). To the extent that including more models and more ensemble members trade off against each other computationally, one participant pointed out that prioritizing large ensembles would be important (*climate modelers convening*). They argued this makes sense since users of the ultimate dataset are likely to want to look at extremes, in which case internal variability is not an insignificant factor (*GEOMIP and climate modelers convening*). Another participant advocated for demonstrating uncertainty by using one method to downscale outputs from one or more models with large ensembles, and choosing fewer models and ensembles to explore scenario and downscaling method uncertainty (*climate modelers convening*).

We decided to downscale the data from all the publicly available models and ensembles that have run the G6-1.5K scenario — MIROC, E3SM, UKESM, and CESM2(WACCM). However, we were unable to access the E3SM data on a timeline that allowed us to include it in the project, and discovered a mismatch in historical and future runs for MIROC that precluded effective statistical downscaling. We therefore downscaled outputs from UKESM and CESM2(WACCM).

03 — Which variables to downscale

Different impact modelers — and different communities — care about SAI’s potential effects on different aspects of the climate system. In practical terms, this means thoughtfully selecting which climate model variables to downscale.

Participants noted that the demand for specific downscaled variables is closely linked to the requirements of particular impacts modeling communities and the tools they already use (*impact modelers, social scientists, and technical experts convenings*). Communities of practice discussed across convenings included hydrologists, glaciologists, crop modelers, public health researchers, ecologists, and economists. Interestingly, one participant noted that in some fields such as economics, higher-resolution climate data may offer limited added value because their analyses are constrained by the coarser resolution of available socio-economic data from statistical agencies (*impact modelers convening*). There was also substantial discussion about the importance of downscaling solar radiation variables directly, rather than deriving it from other downscaled variables (*downscaling experts convening*). Finally, participants recommended focusing the initial data release on the most commonly used variables — such

as temperature, precipitation, humidity, and wind — but designing the downscaling infrastructure to be extensible to more specialized needs (*GEOMIP conference*).

Based on the discussion, we decided to target crop modelers as an important user group, and choose a minimum set of variables to downscale accordingly. From an analysis of crop model inputs, we decided to downscale: near-surface air temperature (maximum, minimum, and mean), precipitation, and surface downwelling solar radiation (rsds). Relative humidity (hurs) was identified as a valuable next variable to downscale for crop models, but was not included in this project due to capacity limitations.

04 — Which bias correction and downscaling algorithm to use

The core technical work of the project involves implementing an algorithm that transforms coarse climate model outputs ($1^\circ \times 1^\circ$) into finer-scale data ($0.25^\circ \times 0.25^\circ$) by comparing them with historical observations. There are many approaches to downscaling, each with significant trade-offs. Based on project constraints (including computation, time, and team expertise), we decided from the outset that dynamical downscaling methods were out of scope, and entered the convenings seeking feedback around statistical downscaling methods specifically.

There was broad consensus across convenings that there is no single “correct” downscaling method, and that implementing multiple methods could help illustrate the uncertainty associated with these methodological choices (*GEOMIP, downscaling experts, climate modelers, and impact modelers convenings*). Many participants agreed that it would be helpful to start with a known method, such as BCSD or ISIMIP (*downscaling experts and technical experts convenings*). However, a key tension is that the well-known and widely used methods also have clear shortcomings and limitations (*downscaling experts convening*) — which is a problem in and of itself, and could also mean these methods are vulnerable to becoming obsolete. One participant pointed out that the act of creating a large global dataset itself could normalize the use of a new and improved downscaling method (*downscaling experts convening*). Several participants noted the limitations of statistical downscaling to capture future relationships between key climate variables in SRM scenarios (*downscaling experts, climate modelers, and technical experts convenings*), particularly in regions with sparse historical data coverage.

Based on these conversations, it was clear that whatever downscaling method we implemented would open new opportunities for research and come with significant limitations. We decided to implement BCSD as a well-understood starting place, as well as a bespoke downscaling algorithm that used quantile delta mapping instead of quantile mapping. The two

methods result in some meaningful differences in the downscaled data, illustrating the impact of method choice. We additionally designed our codebase to make it easy to extend to other downscaling methods.

05 — Which observational dataset to use

A central goal of this project is to enable researchers around the world to engage in studying the potential impacts of SAI. A key challenge is that the Global South — and other important regions of the world, like the Arctic — has fewer observational datasets spanning a shorter historical period. Because we planned to leverage a statistical downscaling method, the quality of the observational baseline was very important for the quality of the outputs. This constrained our choice of observational dataset: it needed to be globally available, consistently gridded, and sufficiently long and complete to support bias-correction and downscaling in all regions.

Many participants noted that the short duration and sparse coverage of observational data in regions such as the Global South pose a challenge, and that there are few ideal options (*civil society, downscaling experts, and climate modelers convenings*). ERA5 was frequently mentioned as a potential observational dataset (*downscaling experts, climate modelers, and technical experts convenings*). One participant suggested using local weather station data (*climate modelers convening*), which could improve data quality, but would introduce significant implementation complexities. Participants also emphasized that the choice of observational dataset matters for comparing downscaled SRM scenarios with other downscaled climate scenarios (*downscaling experts convening*); for example, a direct comparison with NASA NEX data requires matching both the downscaling method (BCSD) and the observational dataset (GMFD, rather than ERA5). Finally, participants highlighted that observational data should be treated as a source of uncertainty (*technical experts convening*), and can limit which variables are feasible to downscale — potentially excluding some variables of interest for understanding SRM impacts, such as droplet number concentration (*climate modelers convening*).

We decided to train our downscaling algorithms on the ERA5 reanalysis dataset, which is globally available. The combination of the BCSD method and ERA5 enables comparison with the World Bank's [downscaled dataset](#).

06 — How to effectively communicate uncertainties

Every technical step in the downscaling process — from scenario design, to running models, to implementing a downscaling method — introduces uncertainties that should influence how the downscaled data is used and interpreted. It's critical to ensure that uncertainties are passed all the way through the chain of experts, and are communicated clearly to end-users. That

includes both the intended users (i.e., researchers who are studying SRM impacts, particularly in the Global South) and other types of users who end up accessing the dataset.

There was strong consensus across convenings that there was no way to avoid serious uncertainties, and that it would be important to both show those uncertainties within the structure of the dataset and tell users about them explicitly (*GEOMIP, civil society, downscaling experts, climate modelers, and social scientists convenings*). Participants in the impact modelers convening especially emphasized the benefit of the structure of the data conveying uncertainties, pointing out that even if a number comes with many caveats, impact modelers will use the data if it's their only option. However, if the data itself demonstrates the uncertainties, like the choice of SRM scenario, they're more likely to grapple with that uncertainty (*impact modeler convenings*). It was emphasized that we will not have ultimate control over who accesses the data and whether they've read what we write — another reason that structurally conveying uncertainty is important (*civil society convening*). Participants also emphasized the utility of documentation of uncertainties in both technical and lay terms (*impact modelers convening*).

Recognizing that only some uncertainties will be captured in the structure of our dataset, we prioritized summarizing uncertainties and dataset limitations in multiple forms, including in the written explainers accompanying the dataset and the detailed methods document. We also created a system of quality assurance flags embedded in the data structure that indicate where the downscaling approaches result in known limitations (e.g., trend distortion or unrealistic output values).

07 — How to ensure appropriate access to the downscaled dataset

Ultimately, this downscaling project is only valuable if it enables a broader community of researchers to more effectively study the global impacts of SAI, and grapple with the benefits and limitations of downscaling with more nuance. That means the dataset must be easy to access and use for researchers around the world, regardless of their technical resources or workflows. Because the dataset will be a public product, and therefore outside our control once released, we also have a responsibility to be intentional in our design and documentation choices to encourage appropriate use and help prevent misinterpretation.

Participants emphasized that for the data to be usable, it has to align with impact researchers' real workflows — including the variables and data formats they regularly use (*impact modelers convening*). There was also an emphasis on building data access tools, like jupyter notebooks, to help a broader audience actually get their hands on the data for further validation and

impacts research (*GEOMIP*). Participants in the civil society convening emphasized the role of active outreach as a key part of accessibility, so a broader community knows both that the data exists, and is likely to use it, and the role of interactive maps in making the data broadly accessible (*civil society convening*). On the flip side, some participants warned against making the data too accessible, recognizing that not all users are likely to take the time to understand or navigate the uncertainties of the downscaled dataset. Some additionally raised the possibility of introducing some sort of more formal access limitation (*climate modelers convening*).

We decided our primary accessibility focus would be building out tooling to help researchers either subset the global dataset to specific regions of interest, or transform the dataset (e.g., units, timesteps, or data formats) to easily fit into their workflows. We aimed to design tooling that's easily usable by all, regardless of coding background or internet bandwidth. We also decided *against* building an interactive map to explore the data, an option we considered prior to the engagements. On net, we felt this would be more likely to encourage use cases that sidestepped the uncertainties and limitations of the dataset.

Other learnings

There were two broader themes that did not directly translate into a project decision, but felt important to surface, as they shaped our thinking about how to share the dataset and judge its impact over the long-term.

First, there was disagreement across the convenings about the utility of encouraging impact analysis that clarifies who “loses” in an SRM scenario. In general, participants in the social scientists convening thought it would be constructive to make clear who would bear the burden of localized SRM downsides. However, several participants in the civil society convening raised concerns, particularly about putting impacted communities under pressure. They pointed out that making this information explicit could make vulnerable communities a target site for stakeholder engagement — by both SAI boosters and skeptics.

Second, there was a shared discomfort across the convenings about the disconnect between providing better, localized information about risk, and the reality (and ambiguity) around who will have ultimate decision-making power. Participants in both the social scientist and civil society convenings discussed how unlikely it seems, given current politics, that SAI research and deployment decisions will be responsive to the public. Civil society participants, for example, raised the question of what good it does for a local community to understand that they face a specific risk if deployment decisions ultimately happen on national or international levels. However, in the social scientist convening, participants also discussed how much

influence the scientific and research community currently has to shape narrative and assumptions in resources that future policymakers will turn to.

Summary

The following table summarizes how our implementation choices map onto the recommendations we heard in the convenings.

Category	Specific recommendation	Implementation
Scenarios	Downscale multiple scenarios to illustrate uncertainty.	Satisfied — Kept G6-1.5K and added a single, purpose-built termination shock scenario.
	Choose scenarios with maximal differences to highlight design uncertainty.	Partially satisfied — Chose to add termination shock instead of diversity of deployment scenarios.
	Prioritize non-ideal scenarios.	Satisfied — Designed, ran, and downscaled a termination shock scenario extending G6-1.5K.
	Consider G6-1.5, GLENS, ARISE-SAI-1.5, and HiLLa specifically.	Partially satisfied — Kept G6-1.5K.
Models and ensembles	Use multiple models and ensemble members to illustrate uncertainty.	Satisfied — Considered all four models running the G6-1.5K scenario; used two models with sufficient public data to enable statistical downscaling, and all available ensemble members.
	Prioritize large ensembles to capture internal variability and support evaluation of extremes.	Satisfied — Addressed insofar as all available ensembles were included.
Variables	Align variables with specific impact-modeling communities' needs.	Satisfied — Targeted crop modelers as the anchor user group, and derived the variable set from common crop-model inputs.
	Consider downscaling temperature, precipitation, humidity, wind, surface ozone, runoff, evapotranspiration, droplet number concentration, soil moisture, ocean variables,	Partially satisfied — Included temperature (max/min/mean), precipitation, near-surface relative humidity, and surface downwelling shortwave radiation.

	and vertical atmospheric profiles	
	Characterize outputs via metrics communities already use (e.g., growing degree days).	Not satisfied — Core output dataset includes only base meteorological variables.
Downscaling method	Implement multiple downscaling methods to illustrate methodological uncertainty.	Satisfied — Included two methods in the initial release, and designed the codebase to extend to additional methods.
	Start from a known, established method (BCSD or ISIMIP).	Satisfied — Implemented BCSD as a well-understood starting point; validated implementation against NEX-GDDP implementation.
	Use a newer, potentially more robust method (e.g., LOCA 2).	Partially satisfied — Implemented a bespoke method which integrates the newer QDM bias-adjustment approach but shares the spatial disaggregation approach of BCSD.
Spatial and temporal resolution	Produce data finer than 0.25 degree resolution.	Not satisfied — Only select researchers requested data at a finer spatial scale, which did not justify using a lesser known training dataset than ERA5 or sacrificing interoperability with NEX-GDDP.
	Produce sub-daily data.	Not satisfied — Only select researchers requested sub-daily data and it would have exceeded the ability of the chosen method, as well as computational limits.
Observational dataset	Use ERA5.	Satisfied — Downscaling algorithm trained on ERA5.
	Use local weather station data.	Not satisfied — Not adopted due to implementation complexity and global consistency requirement.
	Match observational dataset choice for comparability with other downscaled products.	Partially satisfied — By choosing ERA5, sacrificed parity with NASA's downscaled dataset, but achieved it with the World Bank's dataset.
Communicating uncertainty	Both show uncertainty structurally and tell users explicitly.	Satisfied — Structural representation of uncertainty via downscaling multiple models, ensembles, and scenarios; written products all emphasize

		uncertainty; some dimensions of uncertainty not illustrated structurally (e.g., observational dataset) but discussed.
	Documentation in both technical and lay terms.	Not satisfied — Methods and uncertainties are communicated in multiple places and forms, but all with target researchers as the primary audience, with the intention of avoiding data use that sidesteps uncertainties.
Access	Align with researchers' real workflows (formats, variables).	Satisfied — Built access tools to subset and transform downscaled data (units, timesteps, formats).
	Design access that accounts for bandwidth constraints.	Satisfied — Built access tools with bandwidth constraints in mind.
	Active outreach + interactive maps for accessibility	Partially satisfied — Outreach pursued; interactive map declined to avoid encouraging use that sidesteps uncertainties.
	Introduce a more formal access limitation.	Not satisfied — Dataset kept fully public to maintain openness and full accessibility.
	Ensure good archiving and discoverability.	Satisfied — Downscaled datasets are stored in the versioned Icechunk format and published on Source-Coop, which allows for free, high-performance, direct cloud access and long-term archiving.

Acknowledgements

The project and convenings were enabled by dedicated project funding from the Bernard and Anne Spitzer Charitable Trust and The Navigation Fund. The convening recordings transcribed on a local machine using MacWhisper. Freya Chay (CarbonPlan) wrote the first draft of this document. The summary was reviewed by the full project team (CarbonPlan, Cornell, and the Alliance for Just Deliberation on Solar Geoengineering).